# Narendhar T S

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## Education

**Indian Institute of Information Technology, Design and Manufacturing** — Chennai, IN
*B.Tech in Electronics and Communication Engineering (GPA: 8.27)* — 2024 – Expected 2028

**Padma Seshadri Bala Bhavan Senior Secondary School** — Chennai, IN
*CBSE – Class XII: 94.8%, Class X: 96.0%* — 2012 – 2024

## Experience

**Software Development Engineer Intern – Random Walk AI** — May 2026 – Present

- Developed a pipeline reconstructing **interactive 3D scenes** from monocular images using **AI-based depth estimation**.
- Built a **GPU-accelerated renderer** with **off-axis projection** and **real-time head tracking**, synthesizing perspective-correct motion parallax for immersive spatial viewing.

## Projects

**Apollo Bot** | *Acquired* | *Python, Node.js, PostgreSQL, Redis, Linux* — 2021 – 2025

- Built and scaled a distributed Discord bot platform serving **450,000+ communities**, processing **1,000+ messages/second** at peak load.
- Cut request latency by **∼40%** through optimized **PostgreSQL** schemas and **Redis** caching across **500+ concurrent event streams**.
- Architected a multi-process **IPC-based worker system** enabling horizontal scaling across CPU cores.
- Shipped a web dashboard with **OAuth2** authentication for server configuration and analytics.
- Solely owned architecture, deployment, monitoring, and monetization through to a **successful acquisition**.

**Savara 2026 Platform** | *Next.js, React, TypeScript, PostgreSQL* — 2026

- Built and deployed the official ticketing and operations platform for Savara 2026, IIITDM's annual techno-cultural fest, serving **1.3K+ users**, **1.1K participants**, **61 events**, and **1.6K check-ins**.
- Designed **role-based access** for participants, volunteers, and admins, spanning authentication, dashboards, announcements, and event operations.
- Implemented end-to-end **e-ticketing**: ticket issuance, admin dashboard, and automated participation certificate generation.
- Engineered a **QR-based check-in system** with **sub-200ms** verification latency, handling peaks of **478 check-ins/day**.
- Deployed on **DigitalOcean** with **Caddy** and colocated PostgreSQL for fast, reliable on-site ticket scanning.

**ReFile** | *FastAPI, Docker, LangChain, Groq, Supabase* — Oct 2025

- Built an **AI-powered media-processing backend** that turns **natural-language** file requests into executable Linux workflows for video, audio, image, PDF, and **OCR** tasks.
- Implemented **Docker-isolated execution**, ownership checks, sanitized storage, reusable command presets, and **Supabase**-backed file metadata.
- Won **2nd Place** at a **national-level hackathon** conducted by **VIT Chennai**.

**WTFood – College Mess Menu App** | *SvelteKit, PWA* — 2025

- Designed and deployed an **offline-first PWA** used daily by **500+ students**.
- Implemented rule-based meal detection supporting real-time schedules and odd/even week rotations.
- Used **stale-while-revalidate** sync for instant offline access with periodic background updates.

## Achievements & Leadership

- **National Competitions:** Secured **2nd Place** in RetailTech/SaaS and **Top 28 of 750+ teams** at VOID:V1 among **3,000+ participants**.
- **GDSC Coordinator:** Helped organize Vashisht Hackathon 3.0 with **1,200+ participating teams** and **IDFC Bank** sponsorship.
- **Cybersecurity Club Core Member:** Built **Hack n Shield**, a self-hosted attack/defense CTF engine (Python, FastAPI, SQLite) used live at Savara 2026 and Cybersecurity Awareness Month.
- Organized CTFs, quizzes, and security drills for **Cybersecurity Awareness Month**, a Government of India initiative.

## Technical Skills

**Languages**: Python, TypeScript, JavaScript, Rust, C, SQL
**Core CS**: Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Operating Systems, Database Management Systems
**Databases**: PostgreSQL, Redis, SQLite
**Frameworks & Libraries**: Node.js, React, Next.js, Svelte, SvelteKit, Tailwind CSS, Drizzle ORM
**Tools & Platforms**: Linux, Git, GitHub, Docker, Nginx, Cloudflare, Prometheus, Grafana, Supabase, Vercel, Figma